

INTRINSIC IMPEDANCE η

Med. 1 ($z < 0$): incident side. **Med. 2** ($z \geq 0$): transmitted side. Boundary at $z = 0$. (Γ, τ in Table 8-2 below.)

General $\rightarrow$ nonmag. shortcut

$$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \quad (\Omega) \quad \eta_0 = 120\pi \approx 377 \, \Omega$$

Default $\mu_r = 1$ (air, dielectric, “nonmag.”). Only use $\mu_r \neq 1$ if problem says so (“ferromagnetic”, iron, ferrite, or μ_r given explicitly).

Low-loss correction $\sigma/(\omega\epsilon) \ll 1$

$$\eta_c \approx \sqrt{\frac{\mu_r}{\epsilon_r}} \left(1 + j \frac{\sigma}{2\omega\epsilon_r} \right) \quad \text{drop tiny } j \rightarrow \frac{\eta_0}{\sqrt{\epsilon_r}} \quad (\Omega)$$

For Γ/τ just use $\eta_0/\sqrt{\epsilon_r}$. The j term is for accuracy with $|\eta_c|, \theta_\eta$. See LOSSY MEDIA — 5 CASES table for all regimes.

Ê PHASOR TEMPLATES

Wavenumber per medium *nonmag.*

$$k_i = \frac{\omega\sqrt{\epsilon_{r,i}}}{c} \text{ (rad/m)} \Rightarrow k_2 = k_1 \sqrt{\epsilon_{r,2}/\epsilon_{r,1}}$$

Direction table $u \in \{x, y, z\} = \text{axis perp. to boundary}$

Inc dir	Inc	Refl	Trans
$+\hat{u}$	$e^{-jk_1 u}$	$e^{+jk_1 u}$	$e^{-jk_2 u}$
$-\hat{u}$	$e^{+jk_1 u}$	$e^{-jk_1 u}$	$e^{+jk_2 u}$

$+\hat{u}$ direction $\Leftrightarrow$ Med 2 at $u \geq 0$. Reflected sign flips, transmitted matches incident.

General template (incident has pol. $\hat{e}$, amplitude a_0):

$$\mathbf{\tilde{E}}^i = a_0 \hat{e} e^{\mp jk_1 u}, \quad \mathbf{\tilde{E}}^r = \Gamma a_0 \hat{e} e^{\pm jk_1 u}, \quad \mathbf{\tilde{E}}^t = \tau a_0 \hat{e} e^{\mp jk_2 u}$$

(lossy med 2: replace $e^{\mp jk_2 u}$ with $e^{\mp \alpha_2 u} e^{\mp j\beta_2 u}$)

Total in medium 1: $\mathbf{\tilde{E}}_1 = \mathbf{\tilde{E}}^i + \mathbf{\tilde{E}}^r$.

$\hat{e}$ by polarization *plug into templates above*

Pol.	$\hat{e}$	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y = 0$
Lin. $\hat{y}$	$\hat{y}$	$E_x = 0$
Lin. at 45°	$(\hat{x} + \hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x} - j\hat{y}$	$\delta = -\pi/2$
LHC	$\hat{x} + j\hat{y}$	$\delta = +\pi/2$
General	$a_x \hat{x} + a_y e^{j\delta} \hat{y}$	elliptical

Γ, τ act on $\hat{e}$ unchanged. Only swap the polarization, not the formulas.

POWER REFLECTION & TRANSMISSION

% reflected *always*

$$\%P_r = 100 |\Gamma|^2$$

% transmitted *η -ratio matters!*

$$\%P_t = 100 |\tau|^2 \frac{\eta_1}{\eta_2}$$

Check: $\%P_r + \%P_t = 100$.

If medium 2 is denser ($\eta_2 < \eta_1$), $|\tau| > 1$ but $\%P_t < 100\%$ because of the η_1/η_2 factor.

LOSSY MEDIA — 5 CASES

Loss tangent $\epsilon' = \epsilon_r \epsilon_0, \epsilon_0 \approx 1/(36\pi \times 10^9)$

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$$

Type	$\sigma/\omega\epsilon$	η_c, α, β
Lossless ($\sigma=0$)	0	$\eta = \eta_0/\sqrt{\epsilon_r}$ exact; $\alpha = 0, \beta = \omega\sqrt{\epsilon_r}/c$
Low-loss	$< 10^{-2}$	$\eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}} \left(1 + j \frac{\sigma}{2\omega\epsilon_r} \right); \alpha \approx \frac{\sigma\eta_0}{2\sqrt{\epsilon_r}}, \beta \approx \frac{\omega\sqrt{\epsilon_r}}{c}$
Quasi-cond.	$10^{-2} - 10^2$	exact: $(\eta/\sqrt{X})e^{j\theta_\eta}$, see below
Good cond.	$> 10^2$	$\eta_c \approx (1+j)\sqrt{\pi f \mu/\sigma}; \alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}$
Magnetic any	any	keep full $\sqrt{\mu_r/\epsilon_r} \eta_0$

For Γ/τ : drop j correction; use $\eta_0/\sqrt{\epsilon_r}$. Magnetic: keep μ_r .

Exact (quasi-cond) $X = \sqrt{1 + (\epsilon''/\epsilon')^2}$

$$\alpha = \omega \sqrt{\frac{\mu\epsilon'}{2}} \sqrt{X-1}, \beta = \omega \sqrt{\frac{\mu\epsilon'}{2}} \sqrt{X+1}, \theta_\eta = \frac{1}{2} \arctan \frac{\sigma}{\omega\epsilon_r}$$

BOUNDARY

η_1, η_2 — imp. (Ω)
 Γ — refl. coeff (unitless)
 $\tau = 1 + \Gamma$ — trans. coeff
 $k_i = \omega\sqrt{\epsilon_{r,i}}/c$ (rad/m)
 α_2, β_2 — if med 2 lossy
 $z = 0$ — boundary plane
 $\eta_0 = 120\pi \approx 377 \, \Omega$

SNELL / OBLIQUE

θ_i, θ_t — angles from normal
 $n_i = \sqrt{\epsilon_{r,i}}$ — index of refr.
 t_i — slab thickness (m)
 d — lateral disp. (m)
plane of inc. = plane of $\hat{k}$ +normal
 $\perp$: $\mathbf{E} \perp$ this plane
 $\parallel$: $\mathbf{E} \parallel$ this plane

WAVEGUIDE

a, b — wide, narrow dim. (m);
 $a > b$
 m, n — mode indices (int.)
 f_{mn} — cutoff (Hz)
 μ — prop. const (rad/m)
 $Z_{TE/TM}$ — wave imp. (Ω)
 E_x, H_y — field comps (V/m, A/m)
 $\eta = \eta_0/\sqrt{\epsilon_r}$

VELOCITIES

$u_{p0} = c/\sqrt{\epsilon_r}$ — bulk phase vel.
 $u_p = u_{p0}/\sqrt{1 - (f_c/f)^2}$
 $u_g = u_{p0} \sqrt{1 - (f_c/f)^2}$
 $u_p u_g = u_{p0}^2$
 $t = L/u_g$ — travel time
 $c = 3 \times 10^8$ m/s

POWER & UNITS

$\%P_r = 100 |\Gamma|^2$
 $\%P_t = 100 |\tau|^2 \eta_1/\eta_2$
 $\%P_r + \%P_t = 100$
 σ (S/m), ϵ (F/m), μ (H/m)
 $\epsilon_0 = 8.85 \times 10^{-12}$ F/m
 $\mu_0 = 4\pi \times 10^{-7}$ H/m
 $1/(36\pi) \times 10^{-9} \approx \epsilon_0$

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary *angles from normal*

$$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$$

Multilayer chain *stack of slabs* $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$

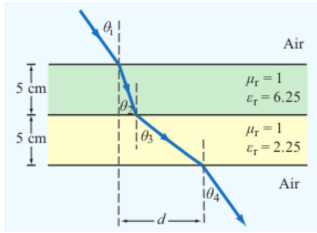
$$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$$

Air–slabs–air shortcut *outer media identical*

$$\theta_{\text{exit}} = \theta_{\text{incident}}$$

Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).

LATERAL DISPLACEMENT



Beam offset through N slabs thickness t_i , refracted angle inside slab i

Slab-indexed θ_i = angle inside slab i

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

Angle-indexed (HW3 P3 style) θ_1 = incident, θ_{i+1} = inside slab i

$$d = \sum_{i=1}^N t_i \tan \theta_{i+1}$$

Same physics, different numbering. In HW3 P3: slab 1 contributes $t_1 \tan \theta_2$, slab 2 contributes $t_2 \tan \theta_3$. The angle in the formula is always the one the ray makes inside that slab.

Air–slab–air: exit angle returns to θ_1 but is laterally shifted by d .

TABLE 8-2 — Γ, τ, R, T SUMMARY

Shorthand: $c_i = \cos \theta_i, c_t = \cos \theta_t$.			
	Normal ($\theta_i = 0$)	$\perp$ pol	$\parallel$ pol
Γ	$\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\frac{\eta_2 c_i - \eta_1 c_t}{\eta_2 c_i + \eta_1 c_t}$	$\frac{\eta_2 c_t - \eta_1 c_i}{\eta_2 c_t + \eta_1 c_i}$
τ	$\frac{2\eta_2}{\eta_2 + \eta_1}$	$\frac{2\eta_2 c_i + \eta_1 c_t}{\eta_2 c_i + \eta_1 c_t}$	$\frac{2\eta_2 c_t + \eta_1 c_i}{\eta_2 c_t + \eta_1 c_i}$
τ rel.	$\tau = 1 + \Gamma$	$\tau_\perp = 1 + \Gamma_\perp$	$\tau_\parallel = (1 + \Gamma_\parallel) \frac{c_i}{c_t}$
R	$ \Gamma ^2$	$ \Gamma_\perp ^2$	$ \Gamma_\parallel ^2$
T	$ \tau ^2 \frac{\eta_1}{\eta_2}$	$ \tau_\perp ^2 \frac{\eta_1 c_t}{\eta_2 c_i}$	$ \tau_\parallel ^2 \frac{\eta_1 c_i}{\eta_2 c_t}$
$R+T$	$T = 1 - R$	$T_\perp = 1 - R_\perp$	$T_\parallel = 1 - R_\parallel$

Notes: $\sin \theta_t = \sqrt{\mu_1 \epsilon_1 / \mu_2 \epsilon_2} \sin \theta_i$; $\eta_i = \sqrt{\mu_i / \epsilon_i}$; nonmag. $\Rightarrow \eta_2/\eta_1 = n_1/n_2$.

RECTANGULAR WAVEGUIDE — MODES

TE mode $E_z = 0, H_z \neq 0$

TM mode $H_z = 0, E_z \neq 0$

Dimensions $a \times b$ with $a > b$ along $\hat{x}, \hat{y}$; wave in $\hat{z}$.

E_x form (TE) *read m, n from arguments*

$$E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$$

coef. on $x = m\pi/a$; coef. on $y = n\pi/b$.

Identify TE_{mn} vs TM_{mn} *cos/sin pattern of E_x is identical for both!*

1. Read the problem statement (e.g. “a TE wave” $\rightarrow$ TE).
2. If $m=0$ or $n=0$ in the mode label $\Rightarrow$ must be TE (TM forbids 0).

3. Source field $H_z(x, y)$ given $\Rightarrow$ TE. Source field $E_z(x, y)$ given $\Rightarrow$ TM.

TE allows at least one of $m, n \geq 1$ (other can be 0). TM needs both $m, n \geq 1$. That's why TE_{10} exists but TM_{10} doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

Shorthand: $C_X = \cos \frac{m\pi x}{a}, S_X = \sin \frac{m\pi x}{a}, C_Y = \cos \frac{n\pi y}{b}, S_Y = \sin \frac{n\pi y}{b}$. All fields carry an $e^{-j\beta z}$ factor (omitted below). $k_c^2 = (m\pi/a)^2 + (n\pi/b)^2$.

TE mode *generator: $\tilde{H}_z$*

$$\tilde{H}_z = H_0 C_X C_Y$$

$$\tilde{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b} \right) H_0 C_X S_Y$$

$$\tilde{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a} \right) H_0 S_X C_Y$$

$$\tilde{E}_z = 0, \quad \tilde{H}_x = -\tilde{E}_y/Z_{TE}, \quad \tilde{H}_y = \tilde{E}_x/Z_{TE}$$

TM mode *generator: $\tilde{E}_z$*

$$\tilde{E}_z = E_0 S_X S_Y$$

$$\tilde{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a} \right) E_0 C_X S_Y$$

$$\tilde{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b} \right) E_0 S_X C_Y$$

$$\tilde{H}_z = 0, \quad \tilde{H}_x = -\tilde{E}_y/Z_{TM}, \quad \tilde{H}_y = \tilde{E}_x/Z_{TM}$$

TEM (plane wave) for reference: $\tilde{E}_x = E_0 e^{-j\beta z}, \tilde{H}_y = \tilde{E}_x/\eta, f_c \text{ N/A}, k = \omega\sqrt{\mu\epsilon}$.

CUTOFF FREQUENCY

f_{mn} m, n integers ≥ 0 (not both zero)

$$f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a} \right)^2 + \left(\frac{n}{b} \right)^2} \text{ (Hz)}$$

Common cutoffs ($a > b$)

Quantity	Formula
u_{p0}	$c/\sqrt{\epsilon_r}$ (hollow: $u_{p0} = c$)
TE_{10}	$u_{p0}/(2a)$ (dominant)
TE_{01}	$u_{p0}/(2b)$
TE_{20}	$u_{p0}/a = 2f_{10}$
$\text{TE}_{11}, \text{TM}_{11}$	$\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$

Propagation requires $f > f_{mn}$.

ϵ_r FROM DISPERSION

Given ω, β , mode (m, n)

$$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a} \right)^2 + \left(\frac{n\pi}{b} \right)^2 \right]$$

Sanity: β in rad/m, a, b in m, ω in rad/s.

WAVE IMPEDANCE & H_y

$$Z_{TE} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}} \quad (\Omega) \quad Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$$
$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \quad H_y = \frac{E_x}{Z_{TE}} \text{ (TE mode)}$$

$Z_{TE} > \eta$ always (denominator < 1); $Z_{TM} < \eta$.

GROUP VELOCITY & TRAVEL TIME

$$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2} \text{ (m/s)}$$
$$u_p = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}} \text{ (m/s)} \Rightarrow u_p u_g = u_{p0}^2 \text{ (m}^2\text{/s}^2\text{)}$$

$t = L/u_g$ (s) travel time through guide of length L
Higher modes have lower $u_g \Rightarrow$ they arrive later. Group delay spread = sum over excited modes.